

Optimization of Multi-Variable Functions

ECON 320 - Introduction to Mathematical Economics

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1. Introduction

Many economic problems require maximizing or minimizing a function of several variables. For example, a firm maximizes profit depending on capital and labor, or a consumer maximizes utility depending on two goods. This is the domain of **multivariable optimization**.

Maximize or minimize $f(x_1, x_2, \dots, x_n)$

possibly **subject to constraints** such as a budget or production function.

2. Unconstrained Optimization in Two Variables

Let

$$z = f(x, y)$$

represent a smooth surface.

2.1 First-Order (Necessary) Conditions

At an interior extremum, small changes in x or y leave z unchanged:

$$f_x = \frac{\partial f}{\partial x} = 0, \quad f_y = \frac{\partial f}{\partial y} = 0$$

The points (x^*, y^*) satisfying these equations are called *stationary points*.

2.2 Second-Order (Sufficient) Conditions

Let

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial x \partial y}$$

and define the **Hessian determinant**

$$D = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx}f_{yy} - (f_{xy})^2$$

Condition	Nature of Stationary Point
$D > 0, f_{xx} < 0$	Local Maximum
$D > 0, f_{xx} > 0$	Local Minimum
$D < 0$	Saddle Point
$D = 0$	Inconclusive

Example 1: Profit Maximization

$$\pi(x, y) = 100x + 80y - 10x^2 - 4y^2 - 4xy$$

$$\pi_x = 100 - 20x - 4y = 0, \quad \pi_y = 80 - 8y - 4x = 0$$

Solve:

$$\begin{cases} 20x + 4y = 100 \\ 4x + 8y = 80 \end{cases} \Rightarrow x = 4, y = 8$$

Second derivatives:

$$f_{xx} = -20, \quad f_{yy} = -8, \quad f_{xy} = -4$$

$$D = (-20)(-8) - (-4)^2 = 144 > 0, \quad f_{xx} < 0 \Rightarrow \text{Local Maximum.}$$

3. Generalization to n Variables

For $f(x_1, \dots, x_n)$:

$$\frac{\partial f}{\partial x_i} = 0, \quad i = 1, 2, \dots, n$$

The second-order conditions use the **Hessian matrix**:

$$H = \begin{bmatrix} f_{x_1x_1} & \cdots & f_{x_1x_n} \\ \vdots & \ddots & \vdots \\ f_{x_nx_1} & \cdots & f_{x_nx_n} \end{bmatrix}$$

To determine the nature of the stationary point, evaluate the **leading principal minors** $D_1, D_2, \dots, D_n$:

Condition	Type of Extremum
Alternating signs ($D_1 < 0, D_2 > 0, D_3 < 0, \dots$)	Maximum
All positive ($D_1 > 0, D_2 > 0, D_3 > 0, \dots$)	Minimum

4. Constrained Optimization: The Lagrange Method

Most economic problems involve constraints.

$$\max_{x,y} f(x,y) \quad \text{subject to} \quad g(x,y) = k$$

4.1 Lagrangian Function

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda(k - g(x, y))$$

4.2 First-Order Conditions (FOCs)

$$\begin{cases} f_x - \lambda g_x = 0 \\ f_y - \lambda g_y = 0 \\ g(x, y) = k \end{cases}$$

At the optimum,

$$\frac{f_x}{g_x} = \frac{f_y}{g_y} = \lambda$$

so the **marginal rates of substitution** are equal to the ratio of the constraint effects.

Example 2: Utility Maximization under a Budget Constraint

$$U(x, y) = x^{0.5}y^{0.5}, \quad p_x x + p_y y = M$$

Lagrangian:

$$\mathcal{L} = x^{0.5}y^{0.5} + \lambda(M - p_x x - p_y y)$$

FOCs:

$$\frac{1}{2}x^{-0.5}y^{0.5} = \lambda p_x, \quad \frac{1}{2}x^{0.5}y^{-0.5} = \lambda p_y$$

Dividing:

$$\frac{y}{x} = \frac{p_x}{p_y} \Rightarrow y = \frac{p_x}{p_y}x$$

Substitute in constraint:

$$p_x x + p_y \left(\frac{p_x}{p_y} x \right) = M \Rightarrow 2p_x x = M \Rightarrow x^* = \frac{M}{2p_x}, \quad y^* = \frac{M}{2p_y}$$

Interpretation: The consumer chooses a point where MRS = price ratio.

5. The Economic Meaning of the Lagrange Multiplier

The multiplier λ represents the **shadow price** or **marginal value of relaxing the constraint**.

$$\frac{\partial f^*}{\partial k} = \lambda$$

In utility maximization, λ equals the marginal utility of income; in production, it represents the marginal output per additional unit of input cost.

6. Multiple Constraints

If there are m constraints:

$$g_i(x_1, \dots, x_n) = k_i, \quad i = 1, \dots, m$$

then

$$\mathcal{L} = f(x_1, \dots, x_n) + \sum_{i=1}^m \lambda_i (k_i - g_i)$$

First-order conditions:

$$f_{x_j} = \sum_{i=1}^m \lambda_i g_{i,x_j}, \quad j = 1, \dots, n$$

7. Second-Order Conditions for Constrained Problems

For a single constraint, the test uses the **bordered Hessian**:

$$H_b = \begin{vmatrix} 0 & g_x & g_y \\ g_x & f_{xx} & f_{xy} \\ g_y & f_{yx} & f_{yy} \end{vmatrix}$$

Sign of H_b	Type of Extremum
$(-1)^m \det(H_b) < 0$	Maximum
$(-1)^m \det(H_b) > 0$	Minimum

Why the $(-1)^m$ factor?

Because each equality constraint adds one border row and column to the Hessian. Each border introduces a sign flip in the determinant pattern.

Hence, the test must account for how many constraints m we have added. That's exactly what the factor $(-1)^m$ does, it adjusts the sign convention.

8. Example 3: Cost Minimization with Cobb–Douglas Technology

$$Q = K^{0.5}L^{0.5}, \quad \text{Minimize } C = wL + rK, \quad Q = \bar{Q}$$

Lagrangian:

$$\mathcal{L} = wL + rK + \lambda(\bar{Q} - K^{0.5}L^{0.5})$$

FOCs:

$$\begin{cases} w - \frac{\lambda}{2}K^{0.5}L^{-0.5} = 0 \\ r - \frac{\lambda}{2}K^{-0.5}L^{0.5} = 0 \end{cases}$$

Dividing:

$$\frac{w}{r} = \frac{K}{L} \Rightarrow \frac{L}{K} = \frac{r}{w}$$

Substitute into constraint:

$$K^* = \bar{Q}\sqrt{\frac{w}{r}}, \quad L^* = \bar{Q}\sqrt{\frac{r}{w}}$$

Interpretation: The firm allocates inputs such that the marginal product per dollar is equalized across inputs.

9. Applications in Economics

Field	Objective Function	Constraint
Consumer Theory	$U(x, y)$	$p_x x + p_y y = M$
Production Theory	$Q(K, L)$	$wL + rK = C$
International Trade	Welfare $W(t)$	Balance of trade constraint
Public Finance	Utility	Government budget constraint
Growth Models	Output or Consumption	Capital accumulation equation